



DMX Tester

Installation & User Manual

Art-Net & sACN Bridge for Raspberry Pi Zero 2 W
Wireless DMX control from iPhone or any mobile browser
RDM Device Discovery & Management

Version 1.0

Table of Contents

3 Installation	5
3.1 Flashing the SD Card	5
3.2 First Boot & SSH Setup	5
3.3 Running the Setup Script	6

4 Network Modes	7
4.1 Show Mode (Hotspot)	7
4.2 Home Mode (Home WiFi)	7

5 Using the App	8
5.1 Opening the App	8
5.2 Faders Screen	8
5.3 Keyboard Screen	9
5.4 Patch Screen	9
5.5 RDM Screen	9
5.6 Settings Screen	10

Overview

DMX Tester turns a Raspberry Pi Zero 2 W into a pocket-sized wireless Art-Net and sACN bridge. Once set up, you control every DMX channel in your rig from the browser on your iPhone — no laptop, no dedicated console, no extra hardware.

The system has two parts that work together:

Part	What it does
Pi Bridge (server)	Receives WebSocket commands from the app and fires Art-Net / sACN UDP packets out over Ethernet to your lighting nodes.
Web App (client)	Runs in Safari (or any browser). Provides faders, a piano-style keyboard, patch editor, RDM discovery, and settings. Can be added to the iPhone home screen as a PWA for full-screen use.

Protocol Support

Protocol	Port	Notes
Art-Net 4	UDP 6454	Broadcast or unicast, up to 32768 universes
sACN (E1.31)	UDP 5568	Multicast or unicast
Art-Net RDM	UDP 6454	Discover, read, and set fixture parameters

NO EXTRA HARDWARE NEEDED

The Pi Zero 2 W has a single WiFi chip, so it either joins your home network or creates its own hotspot — it cannot do both at once. This is handled automatically.

Hardware Requirements

#	Item	Notes
1	Raspberry Pi Zero 2 W	The brain of the system. WiFi + single micro-USB OTG port.
2	MicroSD card	8 GB minimum, Class 10 / A1 recommended. Stores the OS and software.
3	Micro USB → USB-A adapter	OTG adapter. Converts the Pi's data port to a full-size USB socket.
4	USB → Ethernet adapter	Any USB 2.0 gigabit/100M adapter. Plugs into the OTG adapter.
5	Micro USB power supply	5V / 2.5A minimum. Standard phone charger works.
6	Ethernet cable	Connects the adapter to your Art-Net node / lighting console network.
7	iPhone or any browser device	Any device with a modern browser on the same WiFi network.

IMPORTANT: TWO USB PORTS ON PI ZERO 2 W

The Pi Zero 2 W has two micro-USB ports. The **one closer to the centre** is the USB OTG data port — plug your OTG adapter here. The **outer port** is power-only. Use the power port for your charger.

Physical Connection Diagram

```
iPhone / iPad | (WiFi) | [ Raspberry Pi Zero 2 W ] | Ethernet (via USB adapter)
| [ Art-Net Node / DMX Interface ] | DMX 512 (XLR) | [ Lighting Fixtures ]
```

Installation

3.1 Flashing the SD Card

Use **Raspberry Pi Imager** (free, from [raspberrypi.com](https://www.raspberrypi.com)) to write the OS and configure settings before first boot.

- 1 Download Raspberry Pi Imager**
from [raspberrypi.com/software](https://www.raspberrypi.com/software) and install it on your Mac or PC.
- 2 Insert your MicroSD card**
into your computer (use an adapter if needed).
- 3 Choose OS:**
Raspberry Pi OS Lite (64-bit) — no desktop needed.
- 4 Click the gear icon (Advanced Settings)**
and fill in:
 - Hostname: `shtarkair-dmx`
 - Username: `shtarkair` / Password: your choice
 - WiFi SSID: your home network name & password
 - Enable SSH (use password authentication)
- 5 Choose Storage**
→ select your SD card →
Write
.
- 6 Insert the SD card**
into the Pi and power it on.

3.2 First Boot & SSH Setup

Wait ~60 seconds for the Pi to boot. Then, from a Mac terminal on the same network:

```
# Confirm the Pi is online ping shtarkair-dmx.local # Copy your SSH key to the
Pi (avoids password prompts) ssh-copy-id shtarkair@shtarkair-dmx.local # Log in
and allow passwordless sudo (required for setup script) ssh
```

```
shtarkair@shtarkair-dmx.local sudo visudo # Add this line: shtarkair ALL=(ALL) NOPASSWD: ALL
```

HOST KEY WARNING

If you re-flash the SD card and get a "host key changed" error, run:

```
ssh-keygen -R shtarkair-dmx.local
```

3.3 Running the Setup Script

From your Mac, inside the `dmx-tester` project folder:

```
# Install everything on the Pi (run once) bash install-on-pi.sh # Deploy the web app to the Pi bash app/deploy-to-pi.sh shtarkair-dmx.local
```

The setup script will:

- Install Node.js and dependencies
- Install and configure nginx to serve the web app
- Create a systemd service so the bridge starts automatically on boot
- Configure the WiFi hotspot (NetworkManager) with fallback logic

THAT'S IT

After the script completes and the Pi reboots, the bridge is running. No further configuration is needed.

Network Modes

The Pi automatically picks the right mode based on what WiFi networks are in range. You never need to change anything manually.



Show Mode

Home WiFi **not in range**
Pi creates its own hotspot
SSID: **shtarkAir DMXtest**
Password: **dmx12345**
App URL:
http://192.168.4.1

HOTSPOT



Home Mode

Home WiFi **in range**
Pi joins your home network
Connect your phone to
the same home WiFi
App URL:
http://shtarkair-dmx.local

HOME WIFI

PRO TIP — SAVE BOTH AS HOME SCREEN ICONS

In Safari, navigate to each URL and use *Share* → *Add to Home Screen*. Name them "DMX (Show)" and "DMX (Home)" so you always tap the right one.

Switching Modes

The switch is fully automatic and happens during boot:

- Power the Pi **at the venue** — no home WiFi → hotspot starts automatically.
- Power the Pi **at home** — home WiFi found → Pi connects to it as a client.

Using the App

5.1 Opening the App

1 Connect your phone

to the correct WiFi (hotspot at shows, home network at home).

2 Open Safari

and navigate to the URL for your current mode (see Chapter 4).

3 Tap Settings

(bottom tab) → set the Bridge IP → tap Connect.

4 Optional:

Tap Share → Add to Home Screen for a full-screen PWA experience without browser chrome.

5.2 Faders Screen

FADERS

Channel Faders

Each column is one DMX channel (1–512). Drag up/down to set the value 0–255. Tap the value label to type an exact number.

PAGES

Pages

Swipe left/right to switch pages. Each page shows 16 channels. Dots at the bottom show which page you're on.

BLACKOUT

Blackout Button

The red button at the bottom instantly sets all channels to 0. Tap again to restore previous values.

FULL

Full Button

Sets all channels on the current page to 255 (full). Useful for quick checks during focus.

5.3 Keyboard Screen

A piano-style interface that maps keys to DMX channels. Each key lights up when its channel is at full. Tap a key to toggle its channel between 0 and 255. Useful for chases and testing individual dimmers or fixtures.

5.4 Patch Screen

Define named fixtures and map them to DMX addresses. Tap a fixture to expand it and see all its channels. Use this to keep track of what's patched where during a show.

5.5 RDM Screen

RDM (Remote Device Management, ANSI E1.20) lets you read and change fixture settings wirelessly — without walking to each unit.

RDM REQUIREMENTS

Your Art-Net node must support RDM (most modern nodes do). The Pi must be connected via Ethernet to the same network as the node.

Feature	How to use
Discover fixtures	Tap the <i>Discover</i> button. The Pi sends an RDM broadcast and lists all responding fixtures.
View fixture info	Tap any fixture row to open the detail panel: UID, manufacturer, model, DMX footprint, software version.
Set DMX address	In the detail panel, type a new start address (1-512) and tap <i>Set</i> .
Change personality	Tap <i>Load personality names</i> then tap the desired mode to switch the fixture.
Identify fixture	Tap <i>Identify Fixture</i> — the fixture will flash/lamp on until you tap <i>Stop Identify</i> .

5.6 Settings Screen

Setting	Description
Bridge IP	IP address or hostname of the Pi. Use <code>192.168.4.1</code> at shows, <code>shtarkair-dmx.local</code> at home.
Bridge Port	WebSocket port. Default: <code>8080</code> . Do not change unless you modified the server.
Protocol	Art-Net or sACN. Must match what your lighting nodes expect.
Target IP	Destination for DMX packets. Use broadcast (<code>2.255.255.255</code>) to reach all Art-Net nodes, or enter a specific node IP.
Universe	DMX universe number (0-based for Art-Net, 1-based for sACN).

Troubleshooting

Symptom	Likely Cause	Fix
Can't open the app URL	Phone on wrong WiFi, or Pi not booted yet	Wait 45s after power-on. Confirm phone WiFi matches mode (Chapter 4).
App opens but shows "Disconnected"	Wrong Bridge IP in Settings	Go to Settings tab → check Bridge IP → tap Connect.
Fixtures not responding to DMX	Wrong protocol, universe, or Target IP	Check Settings: Protocol, Universe, and Target IP must match your node's config.
RDM Discover returns nothing	Node doesn't support RDM, or Ethernet not connected	Confirm Ethernet cable is plugged in. Check if your Art-Net node supports RDM.
Hotspot SSID not visible	Pi found home WiFi, went into home mode	Move Pi out of home WiFi range and reboot, or connect to home network instead.
SSH: "host key changed"	SD card was re-flashed	Run: <code>ssh-keygen -R shtarkair-dmx.local</code>
App icon on home screen loads wrong network	Saved PWA for wrong mode	Save two separate icons — one per URL (see Chapter 4 tip).

Checking the Bridge Service

SSH into the Pi and run:

```
# Check if the bridge is running sudo systemctl status dmx-bridge # View live
logs sudo journalctl -u dmx-bridge -f # Restart the bridge sudo systemctl
restart dmx-bridge
```

CHAPTER 7

Credits

Shai Shtarker

Creator & Developer

Concept, hardware design, and project vision

shtark.air@gmail.com

github.com/shtarkair/dmx-tester

Built With

Raspberry Pi OS	Linux OS for the Pi bridge
Node.js	WebSocket server runtime
ws	WebSocket server library
Expo / React Native Web	Cross-platform app framework
Zustand	State management
nginx	Web server serving the app on the Pi
NetworkManager (nmcli)	WiFi hotspot management
Art-Net 4 Protocol	DMX over Ethernet (Artistic Licence)
sACN (E1.31)	Streaming ACN DMX protocol (ANSI)
RDM (E1.20)	Remote Device Management protocol (ANSI)

OPEN SOURCE

This project is open source. Contributions, bug reports, and feature requests are welcome at

github.com/shtarkair/dmx-tester